

Sky130 LVS: Required Changes

July 23, 2025

The following changes are required to achieve correct LVS results.

The changes must be applied to the primary Sky130 PDK library: sky130_fd_pr_main

When making CDF changes, be sure to choose “Base” under “CDF Layer” in the Edit CDF form.

Cell Type: all diodes

Change required: CDF

Under Simulation Information, auCDL must be updated:

netlistProcedure: ansCdISubcktCallExtended
componentName: should be empty

Edit CDF (on sjfhw761)

Scope
☐ Library
☒ Cell

CDF Layer
☒ Base
☐ User
☐ Effective

Library: sky130_fd_pr_main
Cell Name: diode_pd2nw_05v5
File Name:
Buttons: Load Save DF Dump

Callback Setup
Form Initialization: PasCdfFormInit
Form Done Procedure:

Simulation Information
Choose List: ☒ By Simulation ☐ By Property
auCdl: Load Data

netlistProcedure: ansCdISubcktCallExtended
componentName:

instParameter: area p j m
termOrder: PLUS MINUS
namePrefix: D
dollarParams:

otherParameter:
propMapping:
modelName: diode_pd2nw_05v5
dollarEqualP:

Cell Type: HV MOS

All 3 HV MOS devices must be updated: nfet_20v0, nfet_20v0_zvt, pfet_20v0

Change required: CDF

Under Simulation Information, auCDL must be updated:

netlistProcedure:	ansCdlSubcktCallExtended
componentName:	should be empty
instParameters:	"nf" added
propMapping:	"nf" added

Edit CDF (on sjfhw761)

Scope
☐ Library
☒ Cell

CDF Layer
☒ Base
☐ User
☐ Effective

Library Name: 130_fd_pr_main File Name:

Cell Name: nfet_20v0

Callback Setup
Form Initialization: PasCdfFormInit Form Done Proc: PasCdfDone

Simulation Information
Choose List: ☒ By Simulation ☐ By Properties auCdl

netlistProcedure	ansCdlSubcktCallExtended	otherParameters	
instParameters	m l w nf	componentName	
termOrder	D G S B	propMapping	nil l gl w tw nf
namePrefix	M	modelName	nfet_20v0
dollarParams		dollarEqualP	

Cell Type: BJT devices

All 3 BJT devices must be updated: npn_05v5 / npn_11v0 / pnp_05v5

Change required: CDF

Under Simulation Information, auCDL must be updated:

netlistProcedure: ansCdlSubcktCallExtended

modelName: *SHOULD MATCH cellname (see example)*

componentName: should be empty

Edit CDF (on sjfhw761)

Scope

☐ Library ☒ **CDF Layer**

☒ Base ☐ User ☐ Effective

Library Name: 130_fd_pr_main File Name: ...

Cell Name: diode_pd2nw_05v5 Load Save DF Dump

Callback Setup

Form Initialization: PasCdfFormInit Form Done Proc:

Component Parameter Simulation Information Interpreted Labels Other Settings

Choose List: ☒ By Simulation ☐ By Property auCdl Load Data

netlistProcedure: ansCdlSubcktCallExtended otherParameter:

instParameters: area p j m componentName:

termOrder: PLUS MINUS propMapping:

namePrefix: D modelName: diode_pd2nw_05v5

dollarParameters: dollarEqualP:

Cell Type: vpp capacitors

Changes are required to all CDFs and some layouts; details below.

All VPP cap edits (CDF and layouts) can be found in the "cap_vpp_fixes.tar" file; un-tar this file inside the directory of the PDK library: sky130_release_0.0.#/libs/sky130_fd_pr_main

CDF Changes

List of cells affected: All "cap_vpp_XXX" cells

Example: cap_vpp_02p4x04p6_m1m2_noshield

Change required:

Under Simulation Information, auCDL "namePrefix" must be "X"

Edit CDF (on sjfhw761)

Scope: ☐ Library ☒ Cell

CDF Layer: ☒ Base ☐ User ☐ Effective

Library: sky130_fd_pr_main File N: ...

Cell Name: m1m2_noshield Load Save DF Dump

Callback Setup: Form Initialization: PasCdfFormInit Form Done Proc: ...

Component Parameter Simulation Information Interpreted Labels Other Settings

Choose List: ☒ By Simulation ☐ By Property auCdl Load Data

netlistProc: ansCdlCompParamPrior otherParameter: ...

instParameter: clw m componentName: 4x04p6_m1m2_noshield

termOrder: c0 c1 b propMapping: ...

namePrefix: X modelName: cap_vpp_02p4x04p6_m1

dollarParams: ... dollarEqualP: ...

Layout Changes

"top" pin must be added to the layouts of some of the cap_vpp cells:

cap_vpp_04p4x04p6_l1m1m2_shieldpo_floatm3
cap_vpp_04p4x04p6_m1m2m3_shieldl1
cap_vpp_04p4x04p6_m1m2m3_shieldl1m5_floatm4
cap_vpp_06p8x06p1_l1m1m2m3_shieldpom4
cap_vpp_06p8x06p1_m1m2m3_shieldl1m4
cap_vpp_08p6x07p8_l1m1m2_shieldpo_floatm3
cap_vpp_08p6x07p8_m1m2m3_shieldl1m5_floatm4
cap_vpp_11p5x11p7_l1m1m2m3m4_shieldm5
cap_vpp_11p5x11p7_l1m1m2m3m4_shieldpom5
cap_vpp_11p5x11p7_l1m1m2m3_shieldm4
cap_vpp_11p5x11p7_l1m1m2m3_shieldpom4
cap_vpp_11p5x11p7_l1m1m2_shieldpom3
cap_vpp_11p5x11p7_m1m2m3m4_shieldl1m5
cap_vpp_11p5x11p7_m1m2m3m4_shieldm5
cap_vpp_11p5x11p7_m1m2m3_shieldl1m5_floatm4

Here is an example for cell “cap_vpp_04p4x04p6_m1m2m3_shieldl1”

